IIM Calcutta

F.I.C.

Finance and Investments club

# Valuation Crasher

# Valuation & Its Importance

Finance and Investments club

## *Why do we value assets/firms?*

- Valuation may be done for several reasons, and depending upon the objective, different numbers can be arrived at for the same entity
- Not because different methods are inconsistent with each other, but because different methods value something different!
- Different purposes require different valuation metrics
- Assets and/or firms are valued for a variety of reasons, some of which have been described below

**Market Investing**

- Various kinds of market investors (mutual funds, common investors, etc.) who are looking to earn money from investing in equity markets
- Such investors rely majorly on publicly available information to perform valuation

**Lending / Financing**

- Banks/NBFCs employ several valuation techniques in order to assess the quality of the collateral they receive for providing a loan
- The cost at which a company can borrow/raise money also depends on its value

**IPOs, Mergers and Acquisitions**

- Companies planning to publicly list their stock (i.e., via an Initial Public Offering) hire Investment Banks in order to value their business that helps determine the listing price
- Companies engaging in M&A deals use valuation to determine terms & conditions of the deal

# Types of Valuation Methods

Finance and Investments club

## *How do we value assets/firms?*

### Discounted Cash Flow (DCF) Methods

- Primary Rationale – the maximum price one should be willing to pay for an asset should not be more than the present value of future expected cash flows one expects to receive
- Different Methods – Dividend Discount Model, Free Cash to Firm (FCFF) based Model, Free Cash to Equity (FCFE) based Model
- Each of the method described above has its pros and cons, and involves a different set of parameters
- Pros- Technically rigorous, allows for sensitivity analysis
- Cons- Highly sensitive to inputs, Prone to overcomplexity

### Relative Valuation Methods

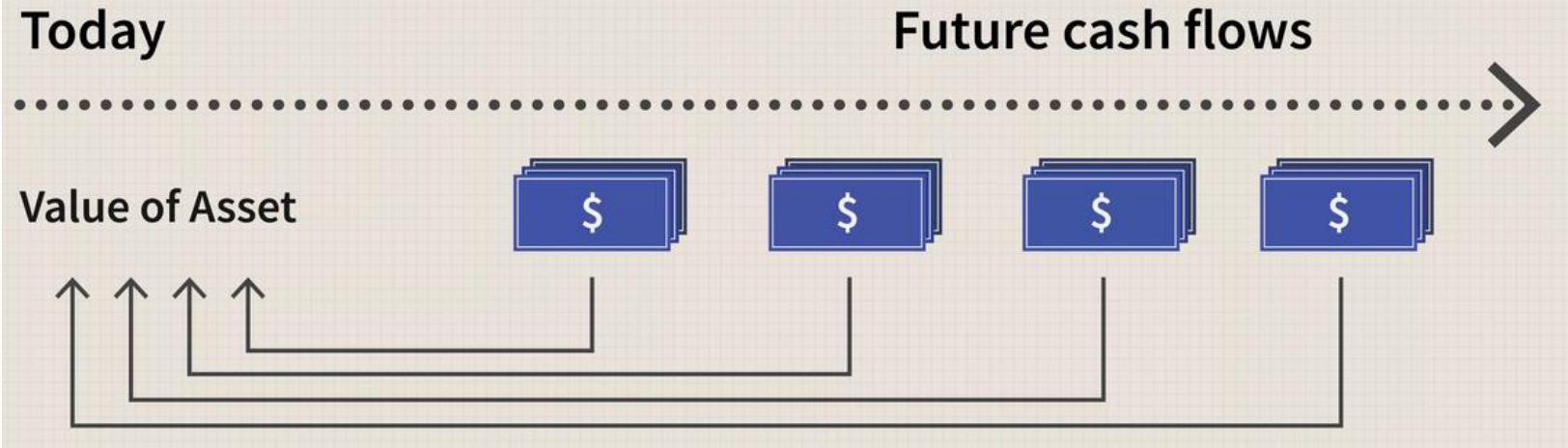
- Primary Rationale – the markets are, on average, efficiently priced, and hence, one can arrive at the price of an asset by looking at the average price the market is willing to pay for similar firms/assets
- Different Methods – Guideline Public Company Method, Guideline Transactions Method, and Prior Transaction Method
- Each of the above methods is used in different circumstances, and can be implemented with different kinds of Multiples
- Pros- Easier to "sell", offers view on current market environment
- Cons- Multiples prone to misuse, assumes market is always right

# Discounted Cash Flow (DCF) Methods

Finance and Investments club

## *Basic Elements*

- Discounting – A cash flow obtained in future is not as valuable as the same amount of cash flow today. Hence, **future cash flows need to be "discounted" in order to arrive at their "present value"**
- The rate at future cash flows are discounted depends upon the risk inherent in the expected future cash flows
- In the context of firm valuation, cash flows need to after-tax
- Projected **Equity Cash Flows are discounted with Cost of Equity** to determine the Equity Value of the common stockholders of a firm
- Projected **Firm Cash Flows are discounted with Weighted Average Cost of Capital (WACC)** to determine the Enterprise Value of a firm

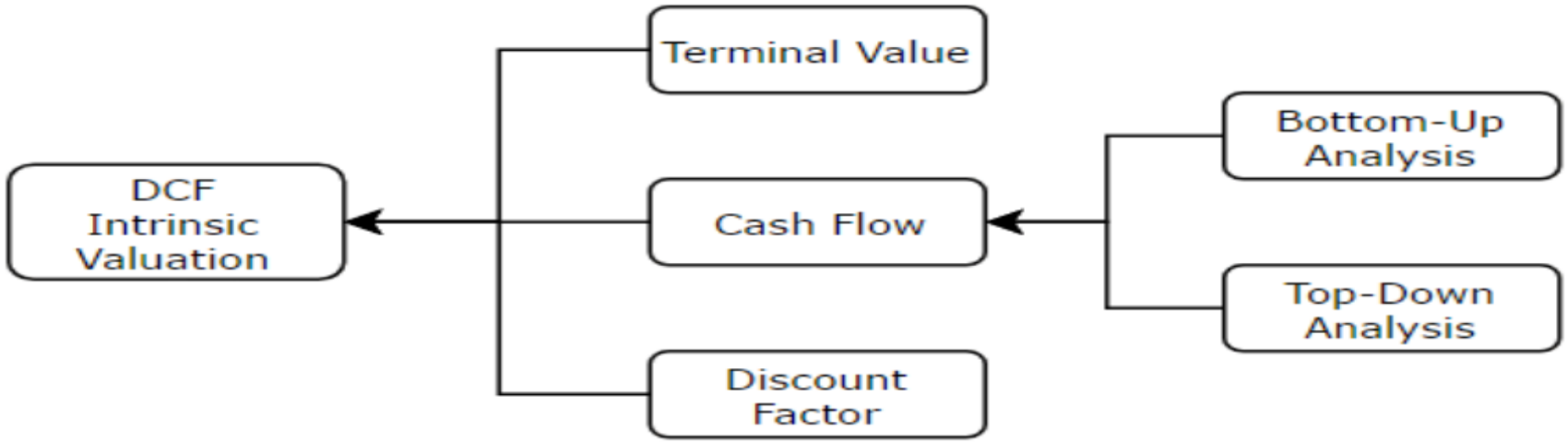



*Source: Investopedia*

# DCF Valuation Inputs

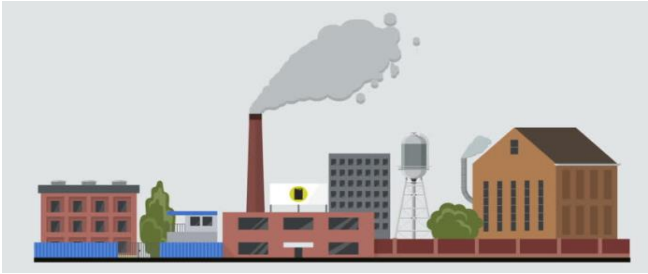
Finance and Investments club

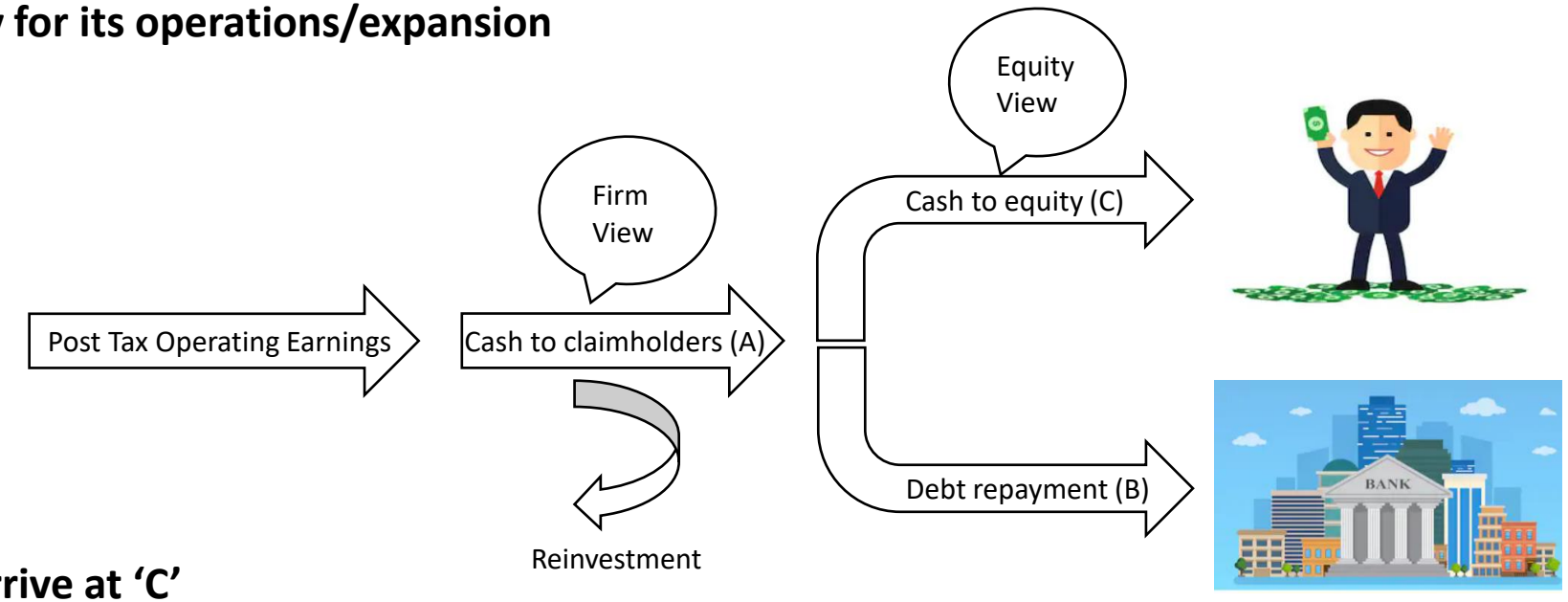
- **Projected Cash Flows**- Cash Flows can be to equity shareholders or to providers of both equity and debt. Usually starts from top line projections and building up from that.
- **Discount rates-** Discount rate should reflect the risk borne by the providers of capital. **Equity cash flows should be discounted at cost of equity and firm cash flows should be discounted at WACC.**
- **Terminal Value-** Reflects the value of equity/firm beyond the forecasted period of cash flows. Assumes that the business will grow at a set growth rate forever beyond the forecast period. **Estimating this growth rate is a critical part of DCF valuation**.



# Which Cash Flows to Project?

- As a **minority investor, we can typically discount the future expected dividends** and arrive at the equity value
- For firms which do not pay dividends, or **for valuing a majority stake, we can discount the free cash flows to arrive at our value**
- The word **"free" means that this cash is available to be distributed to the relevant stakeholders after meeting the reinvestment needs of the company for its operations/expansion**






**2 approaches-**

- **Calculate 'A', subtract 'B' to arrive at 'C'**
- **Calculate 'C' directly**

# Discount rate

Finance and Investments club

The discount rate, or the required rate of return ($r_e$) is the return shareholders demand to compensate them for the **time value of money** tied up in their investment and the **uncertainty of the future cash flows** from these investments.

The **discount rate, should match the nature of cash flows** which are being discounted. Cash flows attributable to equity shareholders are discounted at the cost of equity whereas cash flows attributable to the firm are discounted at Weighted Average Cost of Capital (WACC)

# Estimating discount rates

Finance and Investments club

## *Cost of Equity*

- Risk includes not only less than expected returns but also more than expected returns
- **Diversifiable vs Non Diversifiable risk**- In a diversified portfolio, firm specific risk goes down since (a) very small % of overall portfolio (b) Same risk factor affects different sectors in different ways and risk averages out to zero over time
- However, **non diversifiable risk affects all the sectors in the same direction**
- **Prices of stocks are set by the marginal investor**-most likely to be trading the stock and setting the price. Often large institutions like mutual funds, hedge funds etc. **Marginal Investor is well diversified**

# CAPM

Finance and Investments club

- Why would a marginal investor stop diversifying?
  - Transaction costs leading to marginal costs>marginal gains
  - Active investors believing they can select undervalued stocks via private info
- CAPM Assumptions-
  - 0 transaction costs
  - No access to private info

  **Removed reasons to stop diversifying**
- Result-
  - Investors keep diversifying till they hold the market portfolio
  - Risk of an asset becomes risk added to the market portfolio

# CAPM

Finance and Investments club

CAPM $E(R) = R_f + \beta (R_m - R_f)$

$$\beta_i = \frac{\mathrm{Cov}(r_i, r_m)}{\mathrm{Var}(r_m)}$$

**Slope of regression measures the riskiness of the stock with respect to market value**

# Risk free rate

Finance and Investments club

- Risk free asset-
  - Actual return=Expected return
  - No default risk
  - No reinvestment risk
- Countries that have significant default risk-
  - Estimate default risk (spread) of a country by their bond ratings
  - Look at the largest/safest firms in the country-is their debt traded?

# Equity Risk Premium

Finance and Investments club

- Extra return demanded by investors in shifting their investment from a risk free asset to an asset of average risk
- How to estimate?
  - **Historical averages-** Premium would depend on how far back you take the data, Choice of your risk free security and Arithmetic/Geometric average
  - **Implied equity risk premium**-

Value of market index= ((Expected Dividend next year)/(K(e)-g))

Equity risk premium= K(e)- R(f)

# Estimating discount rates

Finance and Investments club

*Cost of Debt*

- **Debt which is listed on stock exchanges** – Yield to maturity
- **Debt which is rated but not listed on stock exchanges** – Government bond (for the relevant number of years as per instrument) + Default Spread
- **Neither of the above** – Understand the credit profile and benchmark against peers

# WACC

Finance and Investments club

$$WACC = \frac{E}{E+D} * R_E + \frac{D}{E+D} * R_D * (1-T)$$

The weights E/(E+D) and (D/E+D) are calculated based on **market values**

# DCF Models

Finance and Investments club

## *DDM*

Dividend growth at a constant rate (g): (also known as **Gordon Model**)

$$P_0 = \frac{D_0(1+g)^1}{(1+r_e)^1} + \frac{D_0(1+g)^2}{(1+r_e)^2} + \ldots + \frac{D_0(1+g)^\infty}{(1+r_e)^\infty}$$

OR

$$P_0 = D_0 \sum_{t=1}^{\infty} \frac{(1+g)^t}{(1+r_e)^t}$$

OR

$$P_0 = \frac{D_1}{r_e - g}$$

# DCF Models

Finance and Investments club

## *Pros and Cons*

- Pros-
  - Simple
  - Few assumptions
  - Managers often set dividends at sustainable levels. Cash flows are volatile and harder to forecast
- Limitations-
  - Equity claims to built up cash balances are not included
  - Few firms pay consistent dividends
  - Some firms pay higher dividends than cash balances by borrowing/new equity raise which is unsustainable. DDM will overvalue the firm in this case

# DCF Models

Finance and Investments club

## *FCFF Models*

- FCFF= **EBIT*(1-T) + Non Cash charges- Change in Non cash working capital - Capex**
- Discounting FCFF at WACC gives the intrinsic Enterprise Value of the firm
- Enterprise Value= Market Value of Equity + Market Value of Debt + Market value of preferred equity + Market value of minority shares – Cash and Cash equivalents
- Advantages of FCFF models-
  - No need to worry about changing debt structure as FCFF does not consider after debt cash flow
  - WACC less affected by leverage than Cost of equity

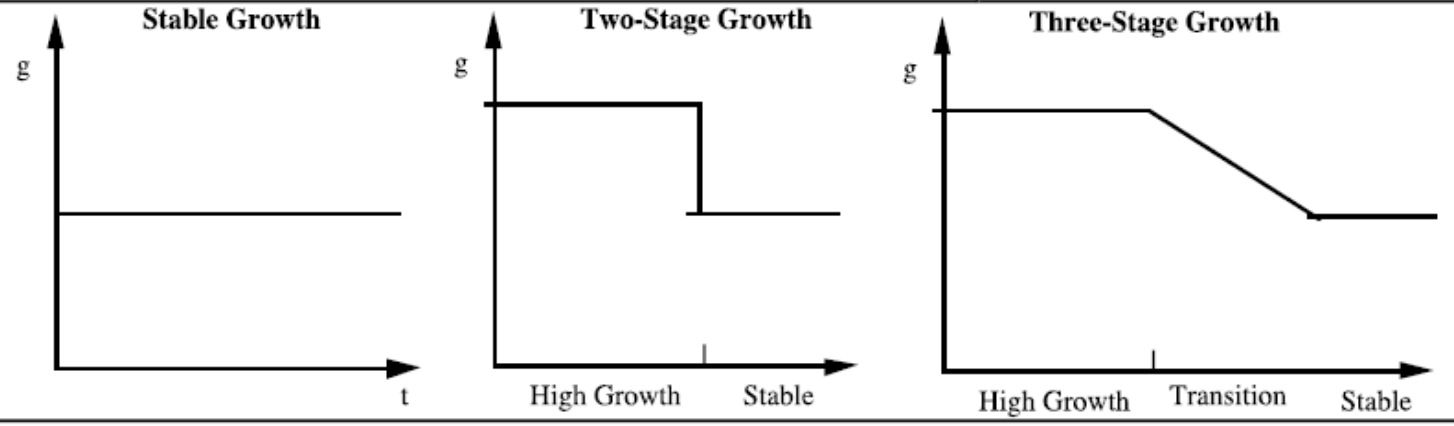
# DCF Models

Finance and Investments club

## *FCFE Models*

- **FCFE can be thought of as potential dividends rather than actual dividends**
- We assume the FCFE will be paid out to shareholders. Consequences-
  - No cash buildup in the firm
  - As a result of the above point, expected growth in FCFE will be due to growth in income from operating assets and not from marketable securities (non core activities)
- FCFE= **FCFF – Interest Expense*(1-T) + Net Borrowings**
- Discounting FCFE at required return on equity gives the intrinsic equity value of the company
- Using an **FCFE** model requires one to **estimate** the **net borrowings** of a firm. In the absence of a plan by the firm under consideration, **estimating net borrowings is extremely subjective** and difficult. In such a case, it is more prudent to use FCFF based model to estimate the Enterprise Value first, and then arrive at Equity Value

# DCF Models

| Choose a | | | |
|---|---|---|---|
| Cash Flow | *Dividends*<br>Expected Dividends to Stockholders | *Cashflows to Equity*<br>Net Income<br>- (1- δ) (Capital Exp. - Deprec'n)<br>- (1- δ) Change in Work. Capital<br>= Free Cash flow to Equity (FCFE)<br>[δ = Debt Ratio] | *Cashflows to Firm*<br>EBIT (1- tax rate)<br>- (Capital Exp. - Deprec'n)<br>- Change in Work. Capital<br>= Free Cash flow to Firm (FCFF) |
| & A Discount Rate | *Cost of Equity*<br>• *Basis*: The riskier the investment, the greater is the cost of equity.<br>• *Models:*<br>CAPM: Riskfree Rate + Beta (Risk Premium)<br>APM: Riskfree Rate + Σ $Beta_j$ ($Risk\ Premium_j$): *n factors* | | *Cost of Capital*<br>WACC = $k_e$ ( E/ (D+E))<br>+ $k_d$ ( D/(D+E))<br>$k_d$ = Current Borrowing Rate (1-t)<br>E,D: Mkt Val of Equity and Debt |
| & a growth pattern |  | | |

# Terminal Value

- Terminal value is the estimated value of a business beyond the explicit forecast period
- Critical part of the financial model, as it typically makes up a large percentage of the total value of a business.

**Terminal Value**

- **Liquidation Value** → Most useful when assets are separable and marketable
- **Multiple Approach** → Easiest approach but makes the valuation a relative valuation
- **Stable Growth Model** → Technically soundest, but requires that you make judgments about when the firm will grow at a stable rate which it can sustain forever, and the excess returns (if any) that it will earn during the period.

$$TV = \frac{[FCFn \times (1 + g)]}{(WACC - g)}$$

# Growth Rate

Finance and Investments club

## *Where does g come from?*

- **g= Retention Ratio * Return on Retained Earnings**
  - Retention Ratio= 1-Dividend Payout Ratio=% of retained earnings not paid out as dividends
- **Duration of growth phase**
  - Young, high growth firms with great potential- longer, say ~5-10 years
  - Large firms in mature, stable industries- shorter, say <=5 years
- **Limits on growth**
  - The stable growth rate cannot exceed the growth rate of the economy
  - Think carefully about the stable growth rate as it is the single most important input which drives the value. Is the terminal return on retained earnings greater than the cost of capital?
  - If yes, does the firm have a competitive advantage over similar firms to justify this assumption? Usually the return on retained earnings converges to industry average during stable growth phase as industry matures and becomes more competitive

# Walk me though a DCF

Finance and Investments club

- **Forecast the free cash flows** to steady state- Usually forecast the top line and build up the other line items from that
- **Sense check** your forecasted cash flows- Ensure steady state assumptions are met (ROC should remain stable, have a good reason if ROC>WACC in steady state etc.)
- **Calculate the discount rate** (Usually WACC if we are discounting FCFF)
- **Calculate Terminal Value**
- **Discount** the cash flows and the terminal value to present
- **Calculate the Equity Value** from Enterprise Value using the bridge
- **Calculated the implied share price**

# Prep Checklist

Finance and Investments club

- **Technicals-**
  - **CFRA concepts should be clear**
  - Essential ratios (profitability, liquidity, efficiency etc.) used to evaluate performance of a business
- **Valuation Methods-** DCF, Relative Valuation
- **Industry Prep-**
  - **Current landscape**- major players, recent developments, current trends
  - **Important performance parameters** (operating and financial metrics) used in the Industry
  - **Relative valuation multiples used in the industry** - Which multiples are used, and why they are used
  - Regulations existing in the industry; recent changes, if any
  - **Recent major deals in the industry** - not just knowledge about deal case facts, must develop your own opinion about the deal (who got the better deal, your opinion about the value, how it affects the industry)
- **Stock Pitch**-
  - Develop a clear opinion on its value
  - Follow news events about it, about its leadership, major changes
  - Outlook and valuation on the stock (how you think it will perform and whether it is a good buy currently)
- **Company Specific Prep**-
  - Recent news and deals of the bank for which you have been shortlisted
  - Formulate an opinion of the deal, and you may discuss with buddy calls whether the valuation is justified. You may also ask questions about the deal if you have doubts or are unable to understand something